

Decision Theory

Solution Sheet 3 and 4

Sheet 3: Admissibility and Bayes Rules

Problem 1: Admissibility under Squared Error Loss

1. Show an admissible estimator T^* is R-incomparable to constant C : If T^* were R-better than C , $R(\theta, T^*) \leq R(\theta, C)$ for all θ . Let $\theta = c$. $R(c, C) = 0$. But $R(c, T^*) = \text{Var}(T^*) + \text{Bias}^2 > 0$ since T^* is non-constant. Thus, T^* cannot be R-better. They are R-incomparable. **2. Show $\delta = T + a$ is inadmissible:** Let $\delta^* = T$. $R(\theta, \delta^*) = \text{Var}(T)$. $R(\theta, \delta) = \text{Var}(T + a) + \text{Bias}(T + a)^2 = \text{Var}(T) + a^2$. Since $\text{Var}(T) < \text{Var}(T) + a^2$, T strictly dominates $T + a$.

Problem 2: Bayes Risk for Crypto

Using values from Sheet 2, Bayes Risk $B(\pi, \delta) = E_\pi[L(\theta, \delta)]$. For $\delta = d_1$, risk is -200. For $\delta = d_2$, risk is -40.

Problem 3: Weighted Absolute Loss

$L(\theta, d) = w(\theta)|\theta - d|$. To find d_{min} , minimize the integral using Leibniz rule:

$$\frac{\partial}{\partial d} \left(\int_{-\infty}^d w(\theta)(d - \theta)\pi(\theta|x)d\theta + \int_d^{\infty} w(\theta)(\theta - d)\pi(\theta|x)d\theta \right) = 0$$

This simplifies to:

$$\int_{-\infty}^d w(\theta)\pi(\theta|x)d\theta = \int_d^{\infty} w(\theta)\pi(\theta|x)d\theta$$

Thus, d_{min} is the median of a density proportional to $w(\theta)\pi(\theta|x)$.

Sheet 4: Risk and Minimax Rules

Problem 1: Normal Distribution Risk

$X|\theta \sim N(\theta, 1)$, $\theta \sim N(0, m)$. $\delta_m(x) = \frac{m}{m+1}x$.

$$\text{Var}(\delta_m) = \frac{m^2}{(m+1)^2}, \quad \text{Bias} = -\frac{\theta}{m+1}$$

$$B(\pi_m, \delta_m) = \int \left(\frac{m^2}{(m+1)^2} + \frac{\theta^2}{(m+1)^2} \right) \pi_m(\theta)d\theta = \frac{m^2 + m}{(m+1)^2} = \frac{m}{m+1}$$

Problem 2 & 3: Minimax and Randomized Rules

Table 1 Analysis: Minimax rule: Find minimum of the maximum row risks. $\min_{\delta}(\max_{\theta} R(\theta, \delta)) = 0.55 = R(\theta, \delta_3)$. Bayes rule for $p = 0.25$: $B(p, \delta_j) = \frac{1}{4}R(\theta_0) + \frac{3}{4}R(\theta_1)$. δ_8 minimizes this to 0.25.

Table 2 Analysis: Show $\delta_* = \frac{3}{49}d_1 + \frac{39}{49}d_2 + \frac{7}{49}d_3$ is Bayes wrt $\pi_* = (\frac{7}{49}, \frac{10}{49}, \frac{32}{49})$.

$$B(\pi_*, d) = \frac{7}{49}R(\theta_1) + \frac{10}{49}R(\theta_2) + \frac{32}{49}R(\theta_3)$$

Evaluating this yields $\frac{81}{49}$ for all p_1, p_2, p_3 . This indicates an equalizer rule, making it a Minimax randomized rule.